Hiram Zuniga

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Summary

Software engineer (B.S. Computer Science, 2026) with hands-on experience building event-driven cloud services on AWS and automating production data pipelines, reporting, and validation systems. Strong CS fundamentals: data structures, algorithms, OOP, parallel computing. Track record of shipping reliable software under real-world constraints, backed by 5 peer-reviewed publications and podium finishes at MIT, Yale, and NASA competitions.

Technical Skills

Languages: Python, SQL, Java (academic), Swift; learning C#

Backend & Cloud: AWS (Lambda, Step Functions, EventBridge, SQS, SNS, S3, DynamoDB, RDS), REST API design and integration, PostgreSQL, event-driven and distributed architectures, Linux

Quality & Automation: unit testing, data validation, automated reporting pipelines, monitoring and alerting, debugging, Git/GitHub, code review, Agile collaboration

Other: machine learning (PyTorch, scikit-learn), Power BI, Power Automate/Apps, Streamlit, LaTeX

Experience

Software Engineering Mentee – Message Router (event-driven system) — Aug 2025 – Present
AWS Mentorship Program, Amazon Web Services — *San Diego, CA*

- Designed and built an event-driven message-routing service composed of decoupled AWS components (EventBridge rules, SNS topics, SQS queues, Lambda functions, Step Functions state machines) backed by PostgreSQL.
- Applied cloud architecture patterns for reliability: dead-letter queues, retries, idempotent consumers, and scheduled event flows; **demoed the system to AWS engineers at the Amazon San Diego office**.
- Extended the platform with a retrieval-augmented (RAG) chatbot built entirely on AWS managed services.

Research Assistant – Software & Systems for Computer Vision — Oct 2025 – Present
CETYS University — *Tijuana, MX*

- Build and ship end-to-end ML systems: designed the real-time, low-latency inference pipeline for an assistive robot integrating camera, LiDAR, and language-model components on constrained hardware.
- Designed reproducible benchmarking suites to evaluate competing model architectures under standardized metrics; work resulted in **5 peer-reviewed papers and manuscripts** (SPIE, KES, Elsevier).
- Own the full lifecycle: data pipelines, training/validation, evaluation, documentation, and demos.

Data Analyst (software automation focus) — Mar 2025 – Sep 2025
I3PL (Cross-Border Logistics Solutions) — *Tijuana, MX*

- Automated daily reporting by integrating multiple REST APIs into scheduled Python pipelines publishing to SharePoint, **cutting manual work by 80%** and standardizing data across sources.
- Built validation rules and monitoring into company-wide data flows, reducing reporting errors and rework; replaced error-prone spreadsheet capture with a validated Power Apps system used across operations.
- Delivered real-time operational dashboards (Power BI, SQL) that reduced container dwell time (>2 hours per load) via live milestone tracking and proactive SLA alerts.

Selected Engineering Projects

Navi – real-time assistive robotics prototype — Mar 2026
HackMerced XI, UC Merced

- Implemented a low-latency perception-to-voice pipeline (YOLO object detection, LiDAR ranging, VLM scene description, LLM voice Q&A) for blind and visually impaired users; designed the inference procedure end to end.

ExoVision – deployed ML application — Oct 2025
NASA International Space Apps Challenge

- Shipped a public web app (exovision.streamlit.app) comparing 6 ML classifiers for exoplanet detection on NASA open data, built and deployed in under 48 hours with a full train/validate/infer pipeline.

Publications & Awards

Author of **5 peer-reviewed papers and manuscripts** (3 first-author), incl. 2 oral presentations at SPIE Optics + Photonics 2026, a paper at KES 2026, and a chapter in revision at *Advances in Computers* (Elsevier). · **3rd place**, MIT iQuHACK 2026 · **3rd place**, Yale YQuantum 2026 · **Regional 1st place & Global Nominee**, NASA Space Apps Challenge 2025 · **2nd place**, GEAR Web3 Hackathon 2023 · NVIDIA Certificate: Getting Started with Deep Learning

Education

CETYS University — 2022 – 2026
B.S. Computer Science Engineering — *Tijuana, MX*

- Coursework: Data Structures, Analysis of Algorithms, OOP, Parallel Computing, Databases, Machine Learning, Deep Learning, Computer Vision, Probability & Statistics. Winter program at Yonsei University, Seoul (2023–2024).
- Languages: Spanish (native), English (TOEFL iBT 100/120).